



$$Y = X\beta + \varepsilon$$

* ~~1. 2.~~ Assumption: Gauss - Markov

1. Linear in Parameter: $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$

2. Random Sampling: $\{(x_{i1}, \dots, x_{ik}, y_i)\}_{i=1}^N$

3. No perfect Collinearity,

4. $E[u | x_1, \dots, x_k] = 0$ Zero Conditional Mean

5. Homoskedasticity 同方差 $E[u^2 | x_1, \dots, x_k] = \sigma^2$

6. Normality, u 与 $x_1, \dots, x_k$ 独立 且 $u \sim N(0, \sigma^2)$

* 1~5 $\rightarrow$ BLUE: Best Linear Unbiased Estimator

性质: $\hat{\beta}$ 在所有 β 中 ($\beta = Cy$) 中, $\hat{\beta}$ 误差最小

$\hat{\beta}$ 是 y 的线性函数 $\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$

$E[\hat{\beta}] = \beta$ (无偏)

只需: 1~4

t6 $\rightarrow$ 参数估计量 e.g. t统计量



Loss Func / Optimize: $y = X\beta + \varepsilon$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$

$$* L(\beta) = \|y - X\beta\|_2^2 = (y - X\beta)^T (y - X\beta)$$

$$\hat{\beta}_{OLS} = \arg \min_{\beta \in \mathbb{R}^p} L(\beta)$$

$$* \nabla_{\beta} L(\beta) = -2X^T (y - X\beta)$$

$$\nabla_{\beta}^2 L(\beta) = 2X^T X \geq 0, \text{ 又若 } X^T X \text{ 正定} \Rightarrow \text{凸}$$

$$\text{令 } \nabla_{\beta} L(\beta) = 0 \rightarrow X^T y = X^T X \hat{\beta}$$

$$\text{若 } X^T X \text{ 可逆 } \text{rank}(X^T X) = p \text{ rank}(X) = p. (X \in \mathbb{R}^{n \times p})$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

* 当 $\varepsilon \sim N(0, \sigma^2 I_n)$, MLE 结果和 OLS 一致:

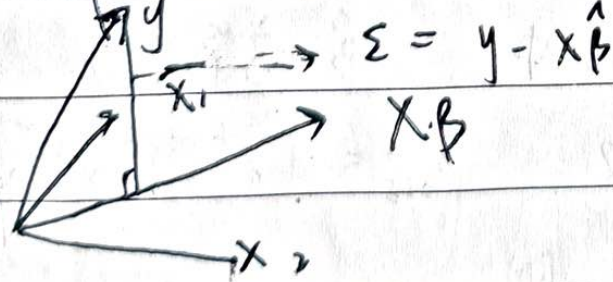
$$\text{Proof: } f(y|X, \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2\right).$$

$$L = \ln f(y_1) \dots \ln f(y_n) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{n}{2\sigma^2} (y - X\beta)^T (y - X\beta)$$

$$\frac{\partial L}{\partial \beta} = -\frac{n}{2\sigma^2} \cdot (-2) \cdot X^T (y - X\beta) = \frac{n}{\sigma^2} X^T (y - X\beta) = 0.$$

$$X^T y = X^T X \hat{\beta} \rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$$

* 几何图: $y = X\beta + \varepsilon \rightarrow$





Σ : heteroskedasticity 异方差.
auto correlation 自相关.

理想: $\text{Var}(\varepsilon|X) = \sigma^2 I_n$, $E[\varepsilon|X] = 0 \Rightarrow \text{理想}: \varepsilon|X \sim N(0, \sigma^2 I_n)$

* $\text{Var}(\varepsilon|X) = \sigma^2 I_n$ 球形扰动项
↳ $E(\varepsilon\varepsilon^T|X) = \sigma^2 I_n$.

当满足球形扰动 spherical disturbance:

有 $\text{Var}(\hat{\beta}|X) = \text{Var}((X^T X)^{-1} X^T y | X)$

又 $\hat{\beta} = (X^T X)^{-1} X^T y = (X^T X)^{-1} X^T (X\beta + \varepsilon) = \beta + (X^T X)^{-1} X^T \varepsilon$

$\Rightarrow \text{Var}(\hat{\beta}|X) = \text{Var}((X^T X)^{-1} X^T \varepsilon | X)$

$= (X^T X)^{-1} X^T \text{Var}(\varepsilon|X) X (X^T X)^{-1}$

$= \sigma^2 (X^T X)^{-1}$

此时, $\text{Var}(\hat{\beta}|X) = \sigma^2 (X^T X)^{-1}$ 是所有线性无偏估计中最小.

$\Rightarrow$ Best. (BLUE).

* 违反 Spherical ...: ①: 异方差 $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \rightarrow (\varepsilon_1^2, \varepsilon_2^2, \varepsilon_3^2)$.

$\rightarrow E[\varepsilon\varepsilon^T|X] = \sigma^2 \Omega$, $\Omega = \begin{bmatrix} w_1 & & \\ & \ddots & \\ & & w_n \end{bmatrix}$

② 自相关: $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \rightarrow (\varepsilon_1, \varepsilon_2, \varepsilon_3)$

$\rightarrow \Omega = \begin{bmatrix} 1 & \dots & \rho_{n-1} \\ \vdots & \ddots & \vdots \\ \rho_{n-1} & \dots & 1 \end{bmatrix}$

* 后果: ①: OLS 无偏但不有效 (非 BLUE)

方法

$\rightarrow$ GLS 重新估计.

② $\text{Var}(\hat{\beta}|X)$ 改变, 普通标准误的假设检验失效.

$\rightarrow$ OLS + robust std error



* GLS:

有 $\text{Var}(\varepsilon|X) = \sigma^2 \Omega$, Ω : symmetric + pos definite.

Ω 可写成 $\Omega^{-1} = C^T C$ (cholesky 分解)

$\Rightarrow$ 对 $y = X\beta + \varepsilon$

变换, $Cy = CX\beta + C\varepsilon$

由 $\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$ 有 $\hat{\beta}_{GLS} = \begin{aligned} &= (CX)^T CX)^{-1} (CX)^T \cdot Cy \\ &= (X^T C^T C X)^{-1} X^T C^T C \cdot y \\ &= (X^T \Omega^{-1} X)^{-1} X^T \Omega^{-1} y \end{aligned}$

实践: Ω 未知 $\leftarrow$ Feasible GLS:

Step 1: Sample data $\rightarrow \hat{\Omega}$

Step 2: GLS: $\hat{\beta}_{FGLS} = (X^T \hat{\Omega}^{-1} X)^{-1} X^T \hat{\Omega}^{-1} y$

* OLS + Robust Std Error

Step 1: $\text{Var}(\hat{\beta}|X) = \cancel{(X^T \Omega^{-1} X)^{-1} X^T \Omega^{-1}} = (X^T X)^{-1} X^T \sigma^2 \Omega X (X^T X)^{-1}$

Sample data $\rightarrow \text{Var}(\varepsilon|X) \rightarrow \hat{\Omega}$ 得 $= \text{Var}(\hat{\beta}|X)$

Step 2: std error



思考: 样本量大

↑

* OLS + Robust Std Error

模型形式:

$$y = \beta_0 + \beta_1 x + u_i$$

$$\hat{\beta}_1 = \frac{\text{cov}(x, y)}{\text{Var}(x)}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\rightarrow (\bar{x}, \bar{y}) \text{ 在 } y = \hat{\beta}_0 + \hat{\beta}_1 x \text{ 上}$$

$$\rightarrow \hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$= \frac{\sum (x_i - \bar{x})(\beta_1 x_i - \beta_1 \bar{x} + u_i)}{\sum (x_i - \bar{x})^2}$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x}$$

$$\hat{\beta}_1 - \beta_1 = \frac{\sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

$$\beta_1 \sum (x_i - \bar{x})(x_i - \bar{x})$$



Q (1): $y \sim (1, x)$. Heteroskedasticity $\rightarrow \hat{\beta}$

$$\hat{\beta} - \beta = (X^T X)^{-1} X^T \varepsilon$$

$$\hat{\beta}_1 - \beta_1 = \frac{\sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

$$\textcircled{1} E[\hat{\beta} | X] = E[\beta + (X^T X)^{-1} X^T \varepsilon | X] = \beta \rightarrow \text{still unbiased}$$

$$\textcircled{2} \text{Var}(\hat{\beta} | X) = \text{Var}[\beta + (X^T X)^{-1} X^T \varepsilon | X] = (X^T X)^{-1} X^T \Omega X (X^T X)^{-1} \\ \rightarrow \neq (X^T X)^{-1} \sigma^2 \\ \rightarrow = (X^T X)^{-1} X^T \sigma^2 D X (X^T X)^{-1} \quad \textcircled{X} \text{ BLUE.}$$

$$\textcircled{3} \hat{\beta}_1 - \beta_1 = \frac{\sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

当 x_i 远离 $\bar{x}$ $= \text{Var}(u_i)$ 的影响会放大 $\rightarrow \text{Var}(\hat{\beta}_1) \uparrow$

Q (2): $y \sim (1, x)$ Autocorrelation $\rightarrow \hat{\beta}$

① still unbiased

② X BLUE

$$\textcircled{3} \hat{\beta}_1 - \beta_1 = \frac{\sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

$$\text{Var}(\hat{\beta}_1 | X) = \frac{\text{Var}(\sum (x_i - \bar{x}) u_i)^2}{S_{xx}^2}$$

~~if X autocorrelation~~

$$= \frac{1}{S_{xx}^2} \left[\sum_i (x_i - \bar{x})^2 \sigma_i^2 + \sum_{i \neq j} (x_i - \bar{x})(x_j - \bar{x}) \text{cov}(u_i, u_j) \right]$$

if $\text{cov}(i, j) > 0$, 且 x_i 有单调趋势 $\Rightarrow \text{Var}(\hat{\beta}_1 | X) \uparrow$ (4 marks)

if $\text{cov}(i, j) < 0$, 且 x_i 有单调趋势 $\Rightarrow \text{Var}(\hat{\beta}_1 | X) \downarrow$ (4 marks)
相邻项有相关性



Q(3): 残差 Auto correlation $+/- \rightarrow$ effective Sample Size 大小.
进而影响 $\hat{\beta}$.

⊗ ESS (effective sample size) ..

当 sample iid $\Rightarrow n$ 充分反映

当 sample 有 correlation / 异方差等问题 $\Rightarrow n_{eff}$

eg. - 1阶自相关:

$$n_{eff} \approx \frac{n}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

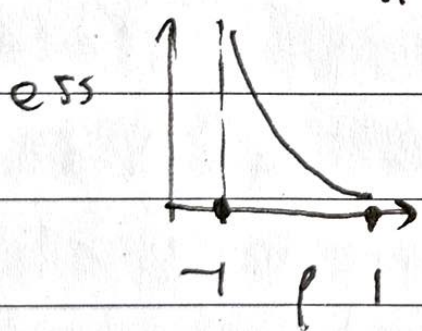
其中: ρ_k 是残差在滞后 k 的 autocorr

if: - 1阶正自相关:

$$n_{eff} \approx n \cdot \frac{1-\rho}{1+\rho}$$

$\Rightarrow \rho > 0$: $\rightarrow n_{eff} < n \rightarrow \text{Var}(\hat{\beta}) \downarrow \rightarrow$ 检验过于乐观

$\Rightarrow \rho < 0$: $\rightarrow n_{eff} > n$ (假) $\rightarrow \text{Var}(\hat{\beta}) \uparrow \rightarrow$ 检验过于悲观



当 $\rho = 1$ 时, $ess = 1$ (并不会无限大)

Q(4): Heteroscedasticity $\rightarrow$ ESS 大小 $\rightarrow \hat{\beta}$

各个样本的方差不同, ESS $\downarrow$

容易低估 $\hat{\beta}$



T 检验 / F 检验

* 证明:

$$\hat{\sigma}^2 = \text{unbiased Estimator} \quad \hat{\sigma}^2 = \frac{SSE}{n-k} = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-k}$$

proof: $\hat{\varepsilon} = y - \hat{y}$

其中: $y = X\beta + \varepsilon$, $\hat{y} = X\hat{\beta} = X \cdot (X^T X)^{-1} X^T y$

$$\rightarrow \hat{\varepsilon} = y - X(X^T X)^{-1} X^T y$$

$$= (I - X(X^T X)^{-1} X^T) y = (I - H)(X\beta + \varepsilon) = (I - H)X\beta + (I - H)\varepsilon$$

$$\therefore H = X(X^T X)^{-1} X^T \quad \hat{\varepsilon} = (I - H)\varepsilon$$

$$\rightarrow \hat{\varepsilon}^T \hat{\varepsilon} = \varepsilon^T (I - H^T) \cdot (I - H) \varepsilon$$

有 $H^T = X \dots = H$

$$\rightarrow \hat{\varepsilon}^T \hat{\varepsilon} = \varepsilon^T (I - H)(I - H) \varepsilon$$

$$\text{有 } (I - H)(I - H) = I - 2H + H^2 = I - 2X(X^T X)^{-1} X^T + X(X^T X)^{-1} X^T = I - H$$

$$\rightarrow \hat{\varepsilon}^T \hat{\varepsilon} = \varepsilon^T (I - H) \varepsilon$$

$$\therefore M := I - H$$

H 有: $H^T = H$, $H^2 = H$, $r(H) = \text{tr}(H) = k$

$$\rightarrow M \text{ 也为对称幂等}, \quad \text{tr}(M) = n - k$$

$$\text{tr}(M) = \text{tr}(I - H) = n - \text{tr}(H) = n - k$$

$$\rightarrow \hat{\varepsilon}^T \hat{\varepsilon} = \varepsilon^T M \varepsilon$$

下面证明: 有 $E[\varepsilon^T \varepsilon] = \sigma^2 I$

$$E[\varepsilon^T A \varepsilon] = \text{tr}(A E[\varepsilon^T \varepsilon]) \quad (\forall A)$$

$$\Rightarrow E[SSE] = \text{tr}(M \sigma^2 I_n)$$

$$= \sigma^2 \text{tr}(M) = \sigma^2 (n - k)$$

$$\Rightarrow E\left[\frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n - k}\right] = \sigma^2 \text{ 是 } \hat{\sigma}^2 \text{ 的无偏}$$



求 $\hat{\sigma}^2$ 的分布:

$$\hat{\sigma}^2 = \frac{SSE}{n-k} = \frac{\sum \varepsilon^2}{n-k} = \frac{\varepsilon^T M \varepsilon}{n-k}$$

由假设: $\varepsilon \sim N(0, \sigma^2 I_n)$

$$\rightarrow \frac{\varepsilon}{\sigma} \sim N(0, I_n)$$

$$\rightarrow \frac{\varepsilon^T}{\sigma} M \frac{\varepsilon}{\sigma} \sim \chi_{n-k}^2 \quad \text{即} \quad \frac{\varepsilon^T M \varepsilon}{\sigma^2} \sim \chi_{n-k}^2 = \frac{\sum \varepsilon^2}{\sigma^2} = \frac{ESS}{\sigma^2}$$

$$= \frac{\hat{\sigma}^2 (n-k)}{\sigma^2} \sim \chi_{n-k}^2$$

proof:

下列正交矩阵: $Z = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$, $z_i \sim N(0, 1) \Rightarrow Z \sim N(0, I)$

$$q = Z^T A Z, \quad A \text{ symmetric}$$

有 For sym. A $\rightarrow A \Rightarrow C \Lambda C^T$, Λ diagonal, C orthogonal.

$$\Rightarrow q = Z^T C \Lambda C^T Z$$

$$\text{又 } Z \sim N(0, I) \Rightarrow C^T Z \sim N(0, C^T I C) = N(0, I)$$

$$\text{令 } y = C^T Z \sim N(0, I) \quad \text{有 } q = y^T \Lambda y$$

$$\text{令 } \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}, \quad \text{有 } q = y^T \Lambda y = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2$$

$$\Rightarrow q \sim \chi_j^2, \quad j \in \{ \lambda_1, \dots, \lambda_n \} \text{ 中 } \lambda_i = 1 \text{ 个数}$$

$\Rightarrow$ 同样的, 对 $\frac{\varepsilon^T}{\sigma} M \frac{\varepsilon}{\sigma}$, M 同 A , 可以分

$$\text{tr}(M) = r(M) = n-k \Rightarrow \frac{\varepsilon^T M \varepsilon}{\sigma^2} \sim \chi_{n-k}^2$$

下证: $\hat{\beta}_i$ 的分布:

$$\text{有 } \beta_0 \sim N(\beta_0, \sigma^2 (X^T X)^{-1})$$

$$\text{对 } \hat{\beta}_i: \hat{\beta}_i \sim N(\beta_i, \sigma^2 (X^T X)^{-1}_{ii})$$

$$\rightarrow \frac{\hat{\beta}_i - \beta_i}{\sqrt{\sigma^2 (X^T X)^{-1}_{ii}}} \sim N(0, 1)$$



* t 检验:

$$T = \frac{\bar{Y}}{\sqrt{\frac{V}{n}}} \sim t_n$$

其中: $Y \sim N(0,1)$

$$V = Z_1^2 + \dots + Z_n^2 \sim \chi_n^2$$

Proof:

$$1/3 \quad Y = \frac{\hat{\beta}_i - \beta_i}{\sqrt{\sigma^2 (X^T X)^{-1}_{ii}}}$$

$$V = \frac{\hat{\sigma}^2 (n-k)}{\sigma^2}$$

$$\text{有 } \frac{\hat{\beta}_i - \beta_i}{\sqrt{\sigma^2 (X^T X)^{-1}_{ii}}} \sim N(0,1)$$

$$\frac{\hat{\sigma}^2 (n-k)}{\sigma^2} \sim \chi_{n-k}^2$$

$$t_i = \frac{\frac{\hat{\beta}_i - \beta_i}{\sqrt{\sigma^2 (X^T X)^{-1}_{ii}}}}{\sqrt{\frac{\hat{\sigma}^2 (n-k)}{\sigma^2}}} \sim t_{n-k}$$

$$t_i = \frac{\hat{\beta}_i - \beta_i}{\sqrt{\hat{\sigma}^2 (X^T X)^{-1}_{ii}}} \sim t_{n-k}$$

其中: $\hat{\sigma}^2 (X^T X)^{-1}_{ii}$ 是 β_i 的 $\text{Var}(\beta_i)$ 的估计 $\hat{\text{Var}}(\beta_i)$

$$\Rightarrow \beta_i \text{ 标准误差} = \sqrt{\hat{\sigma}^2 (X^T X)^{-1}_{ii}}$$

$$\Rightarrow t_i = \frac{\hat{\beta}_i - \beta_i}{\text{s.e.}(\beta_i)} \sim t_{n-k}$$

检验: $H_0: \beta_i = 0$

若 H_0 成立:

$H_1: \beta_i \neq 0$

$$\text{有 } t_i = \frac{\hat{\beta}_i}{\text{s.e.}(\hat{\beta}_i)} \sim t_{n-k}$$



$$X \sim \chi_a^2 \quad Y \sim \chi_b^2$$

$$F = \frac{\frac{X}{a}}{\frac{Y}{b}} \sim F_{a,b}$$

* F :

$$R\hat{\beta} - r = 0.$$

若 $g \sim N(\mu, \Sigma)$ 有 $(g - \mu)^T \Sigma^{-1} (g - \mu) \sim \chi_p^2$

$\Rightarrow$

$$\text{有 } R\hat{\beta} - r \sim N(R\beta - r, R\sigma^2(X^T X)^{-1} R^T)$$

$$\rightarrow \frac{(R\hat{\beta} - r)^T [R(X^T X)^{-1} R^T]^{-1} (R\hat{\beta} - r)}{p} \sim \chi_p^2$$

又有 $\frac{\hat{\sigma}^2(n-k)}{\sigma^2} \sim \chi_{n-k}^2$

$$\Rightarrow F = \frac{(R\hat{\beta} - r)^T [R(X^T X)^{-1} R^T]^{-1} (R\hat{\beta} - r)}{p \cdot \hat{\sigma}^2} \sim F_{p, n-k}$$

$$\hat{\sigma}^2 = \frac{ESS}{n-k}$$

$$\Rightarrow = \frac{(ESS_r - ESS_u)/p}{ESS_u / (n-k)} \sim F_{p, n-k}$$

 R^2

$$SST = SSE + SSR$$

⊗ Adjusted R^2

$$\tilde{R}_{adj} = 1 - (1 - R^2) \frac{n-1}{n-k-1}$$

* Def:

$$R^2 = \frac{SSR}{SST} = \frac{\|\hat{Y} - \bar{Y}\|^2}{\|Y - \bar{Y}\|^2} = \frac{\text{Var}(\hat{Y})}{\text{Var}(Y)}$$

$$= 1 - \frac{SSE}{SST} = 1 - \frac{\|\hat{\varepsilon}\|^2}{\|Y - \bar{Y}\|^2}, \quad \hat{\varepsilon} = Y - \hat{Y} = Y - X\hat{\beta}$$

Q11. 样本复制一倍, R^2 ?

① 对 β 影响:

$$\text{有 } \hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\text{复制一倍} \Rightarrow \begin{pmatrix} X \\ X \end{pmatrix}, \begin{pmatrix} Y \\ Y \end{pmatrix}$$

$$\text{有 } \hat{\beta}' = \left(\begin{pmatrix} X \\ X \end{pmatrix}^T \begin{pmatrix} X \\ X \end{pmatrix} \right)^{-1} \begin{pmatrix} X \\ X \end{pmatrix}^T \begin{pmatrix} Y \\ Y \end{pmatrix} = \cancel{(2X^T X)^{-1}}$$

$$= \left(\begin{pmatrix} X^T & X^T \end{pmatrix} \begin{pmatrix} X \\ X \end{pmatrix} \right)^{-1} \begin{pmatrix} X^T & X^T \end{pmatrix} \begin{pmatrix} Y \\ Y \end{pmatrix}$$

$$= (2X^T X)^{-1} \cdot 2X^T Y = (X^T X)^{-1} X^T Y$$

$$= \hat{\beta}$$

$$\Rightarrow \hat{\beta} \text{ 不变}$$

② $\hat{\beta}$ 不变 $\hat{Y}' = X\hat{\beta}' = \hat{Y}$ $\hat{Y}$ 不变 $\Rightarrow R^2$ 不变

$\text{Var}[(2X^T X)^{-1} (2X^T \varepsilon)] \neq \text{Var}(X^T X)^{-1} X^T \varepsilon$
 样本复制 ≠ 常数乘 2.
 $= \text{Var}(2X^T X)^{-1} X^T \varepsilon + \text{Var}(\dots)$
 $= \frac{1}{2} \dots$



Q14. 样本复制 - 1倍, t ?

$t_i = \frac{\hat{\beta}_i}{\text{s.e.}(\hat{\beta}_i)}$
 有 $\hat{\beta}_i$ 不变
 2倍 s.e. $(\hat{\beta}_i)$

$\text{Var}(\hat{\beta}) = \text{Var}[(X^T X)^{-1} X^T Y] = \text{Var}[(X^T X)^{-1} X^T (X\beta + \varepsilon)]$
 $= \text{Var}(\beta + (X^T X)^{-1} X^T \varepsilon)$
 $= (X^T X)^{-1} X^T \text{Var}(\varepsilon) X (X^T X)^{-1}$

$\Rightarrow \text{Var}(\varepsilon) = \sigma^2 I_n$
 $= \sigma^2 (X^T X)^{-1}$

$\Rightarrow X \rightarrow \begin{pmatrix} X \\ X \end{pmatrix}$

$\text{Var}(\hat{\beta}) = \frac{1}{2} \sigma^2 (X^T X)^{-1} \Rightarrow t \rightarrow \sqrt{2} \frac{t}{\sqrt{2}}$

实际: 无 σ^2 , 一般用 $\hat{\sigma}^2$ unbiased

$\hat{\sigma}^2 = \frac{\text{SSE}}{n-k}$, n : sample, k : feature.
 $= \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-k}$

2 $\hat{\beta}$ 不变, $\hat{\varepsilon}$ 不变, 只是复制 - 1倍.

$\Rightarrow \hat{\sigma}_{\text{new}}^2 = \frac{\begin{pmatrix} \hat{\varepsilon} \\ \hat{\varepsilon} \end{pmatrix}^T \begin{pmatrix} \hat{\varepsilon} \\ \hat{\varepsilon} \end{pmatrix}}{2n-k} = \frac{2 \hat{\varepsilon}^T \hat{\varepsilon}}{2n-k}$

$\Rightarrow \frac{\hat{\sigma}_{\text{new}}^2}{\hat{\sigma}^2} = \frac{2 \hat{\varepsilon}^T \hat{\varepsilon}}{2n-k} \cdot \frac{n-k}{\hat{\varepsilon}^T \hat{\varepsilon}} = \frac{2(n-k)}{2n-k} = 1 - \frac{k}{2n-k}$

$\Rightarrow$ 实际中: t 会 $\frac{t_{\text{new}}}{t} = \sqrt{2 + \frac{k}{n-k}}$ 略大于 $\sqrt{2}$

当 $n \gg k$ 时: $\hat{\sigma}$ 近似不变, $t \rightarrow \sqrt{2}$



Q(3), $X, Y \stackrel{iid}{\sim} N(0, 1)$, $\{x_i\}_{i=1}^n, \{y_i\}_{i=1}^n, y \sim(x), E[k'] = ?$

$$y \sim (1, x) \phi, R^2 = r^2$$

$$\text{又 } X, Y \text{ iid}, \rho = 0$$

$$\text{当 } \rho = 0 \text{ 时: } r^2 \text{ 有 } r^2 \sim \text{Beta}\left(\frac{1}{2}, \frac{n-2}{2}\right)$$

$$\text{对于 } \text{Beta}(a, b) \text{ 有 } E[z] = \frac{a}{a+b}$$

$$\Rightarrow E[r^2] = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{n-2}{2}} = \frac{1}{n-1} = E[k']$$

$$* R^2 = r^2 (y \sim (1, x)) \quad \text{且 } r \text{ 是 } \rho \text{ 的估计}$$

见 P18 件

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

$$\rho = \frac{\text{cov}(X, Y)}{\text{std}(X) \text{std}(Y)}$$

r 的分布 $\left\{ \begin{array}{l} \text{当 } \rho = 0, r^2 \sim \text{Beta}\left(\frac{1}{2}, \frac{n-2}{2}\right), E[r^2] = \frac{1}{n-1} \\ \text{有 } \sigma^2 \text{ 分布: } \hat{\sigma}^2 = \frac{\sum \sum}{n-k} \sim \chi_{n-k}^2 \end{array} \right.$

$$\Rightarrow H_0: \rho = 0 \text{ 的 } t: t = \frac{\hat{\beta} - 0}{\text{s.e.}(\hat{\beta})}$$

$$\text{其中: } \text{s.e.}(\hat{\beta}) = (X^T X)^{-1} \hat{\sigma}$$

$$\text{s.e.}(\hat{\beta}) \text{ 为 } \chi^2 \text{ 分布}$$

$$\text{② } \hat{\beta} \sim (0, \sqrt{(X^T X)^{-1} \sigma^2})$$

Q(4), $x, y \in \mathbb{R}^n$ 一维/二维/三维 $\beta_1 \Rightarrow y = \beta_1 x + \varepsilon, x = \beta_2 y + \varepsilon \Rightarrow \beta_1, \beta_2 \text{ 范围}$

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (y_i - \bar{y})^2}$$

$$\Rightarrow \beta_1 - \beta_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})^2}{\sum (x_i - \bar{x})^2 - \sum (y_i - \bar{y})^2} =$$

无偏估计: $\beta_1 = \frac{\sum x_i y_i}{\sum x_i^2}, \beta_2 = \frac{\sum x_i y_i}{\sum y_i^2}, \beta_1 \cdot \beta_2 = \frac{(\sum x_i y_i)^2}{\sum x_i^2 \cdot \sum y_i^2}$

Cauchy: $(\sum x_i y_i)^2 \leq \sum x_i^2 \cdot \sum y_i^2 \Rightarrow 0 \leq \beta_1 \beta_2 \leq 1$



Q(5): $SST = SSE + SSR$ 成立条件.

$R^2 = R^2$ 会不会负!

[成立条件]: (-元/3元)

① OLS + 截距/变量中心化

$$\Rightarrow SST = SSE + SSR, R^2 = \frac{SSR}{SST}$$

② 无截距/未中心化

$\Rightarrow SST$ 无法分解

$$R^2 \text{ 范围不在 } [0, 1], R^2 = 1 - \frac{SSE}{SST} < 0 / > 1 \text{ 都有可能}$$

[推导]: $SST = \|y - \bar{y} \cdot 1\|^2$ $SSE = \|\hat{y} - y\|^2$ $SSR = \|\hat{y} - \bar{y} \cdot 1\|^2$

有 $y - \bar{y} \cdot 1 = \hat{y} - \bar{y} \cdot 1 + y - \hat{y} = \hat{y} - \bar{y} \cdot 1 + \vec{e}$

$$\|y - \bar{y} \cdot 1\|^2 = \|\hat{y} - \bar{y} \cdot 1\|^2 + \|\vec{e}\|^2 + (\hat{y} - \bar{y} \cdot 1)^T \vec{e}$$

显然: 残差 $\perp$ $\omega_1(x)$

[排除 R^2]: 当 SSE 非负 $\Rightarrow R^2 \leq 1$

Q(6): 遗漏截距项 影响?

①: $\hat{\beta}$ 不再 unbiased: 把截距作用强行分配到 x 上.

②: $\sum \hat{\epsilon}_i \neq 0$.

③: 回归线不经过 $(\bar{x}, \bar{y})$.

④: R^2 可能 < 0 或 > 1

⑤: $\hat{\epsilon} \not\perp \text{COL}(x)$



Q(7) OLS时, 数据量太大, 无法一次读进内存.

$$n \gg p, \quad \hat{\beta} = (X^T X)^{-1} X^T Y, \quad X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

(x_i, y_i) - 批读进内存.

$$\hat{\beta} = \left[\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}^T \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right]^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}^T Y = \left[\sum_{i=1}^n x_i^T x_i \right]^{-1} [x_1^T, \dots, x_n^T] \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$= \left[\sum_{i=1}^n x_i^T x_i \right]^{-1} \left[\sum_{i=1}^n x_i^T y_i \right] \Rightarrow \text{实际只用 } n \text{ 个 } 1 \times 1 \text{ 计算.}$$

$\downarrow \mathbb{R}^{p \times p} \quad \quad \quad \downarrow \mathbb{R}^{p \times 1}$

Q(8) $y \sim (1, x), \quad x \sim (1, y), \quad \beta_1, \beta_2 = R^2, \quad R^2 =$

$$\beta_1 = \frac{\text{cov}(x, y)}{\text{Var}(x)} \quad \beta_2 = \frac{\text{cov}(x, y)}{\text{Var}(y)} \quad \beta_1 \beta_2 = \frac{\text{cov}(x, y)^2}{\text{Var}(x) \text{Var}(y)} = r^2 = R^2$$

相关系数² = 拟合度

Q(9) $y \sim (x)$ 的回归系数 1, $x \sim (y)$ 的回归系数条件?

Q... $\Rightarrow K$

$$\beta_1 = \frac{\sum x_i y_i}{\sum x_i^2} = 1 \quad \beta_2 = \frac{\sum x_i y_i}{\sum y_i^2} \quad \beta_1 \beta_2 = r^2 \quad \text{相关系数}^2$$

$$\cos^2 \theta \leq 1 \Rightarrow \beta_2 \in (0, 1] \quad \text{取 } = 1, \text{ 当 } X=Y$$

$$\beta_1 = |K|, \quad 0 < \beta_1 \beta_2 \leq 1 \Rightarrow \beta_2 \leq \frac{1}{|K|}$$

Q(10): $y \sim (1, x_1, x_2) \Rightarrow R^2, \quad \text{corr}(x_1, y) = \beta_1, \quad \text{corr}(x_2, y) = \beta_2, \quad \text{corr}(\hat{y}, x_1) =$

有 $\boxed{\text{cov}(\hat{y} - y, x_1) = 0} \Rightarrow \text{cov}(\hat{y}, x_1) = \text{cov}(y, x_1)$

$$\Rightarrow \text{corr}(\hat{y}, x_1) = \frac{\text{cov}(\hat{y}, x_1)}{\sqrt{\text{Var}(\hat{y})} \sqrt{\text{Var}(x_1)}} = \frac{\text{cov}(y, x_1)}{\sqrt{\text{Var}(\hat{y})} \sqrt{\text{Var}(x_1)}} = \frac{\text{cov}(y, x_1)}{\sqrt{\text{Var}(y)} \sqrt{\text{Var}(x_1)}} = \frac{\beta_1}{R}$$

Proof: $\hat{Y} = X \hat{\beta} = X (X^T X)^{-1} X^T Y$

同乘 X : $X^T \hat{Y} = X^T Y$

$$\text{或 } X^T (\hat{Y} - Y) = 0 \quad \text{证.}$$



$y \sim (x)$ 的 R^2 和 $\text{corr}(\hat{y}, y)$ 关系?

Q (11) 如果不是 OLS, 而是 $y \sim f(x)$, $y \sim f(x)$, 有什么关系?

$$y \sim (x), \quad \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{x^T y}{x^T x}$$

$$\text{有 } x^T (y - \hat{y}) = 0$$

$$\left\{ \begin{array}{l} \text{由 } \lambda: x^T (y - \hat{\beta} x) = 0 \quad \text{即 } x^T y = x^T \hat{y} = x^T \hat{\beta} x = \hat{\beta} x^T x \\ \text{由 } \hat{\beta}: x^T y = \frac{x^T y}{x^T x} \cdot x^T x \end{array} \right. \quad \#$$

$$x^T y = x^T \hat{y} \quad \text{同乘 } \hat{\beta}:$$

$$\hat{y}^T \hat{y} = y^T \hat{y}$$

无约束 OLS

的正交性等式推导

无约束

$$R_o^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\|y - \hat{y}\|^2}{y^T y}$$

$$\text{其中 } SSE = \|y - \hat{y}\|^2 = y^T y + \hat{y}^T \hat{y} - 2y^T \hat{y}$$

$$\text{又 } \hat{y}^T \hat{y} = y^T \hat{y} \Rightarrow SSE = y^T y - \hat{y}^T \hat{y}$$

$$\Rightarrow R_o^2 = \frac{\hat{y}^T \hat{y}}{y^T y}$$

$$\text{corr}(y, \hat{y}) = \frac{y^T \hat{y}}{\sqrt{y^T y} \cdot \sqrt{\hat{y}^T \hat{y}}} = \frac{y^T \hat{y}}{\sqrt{(y^T y)(\hat{y}^T \hat{y})}} = \frac{y^T \hat{y}}{y^T \hat{y}} = \sqrt{\frac{\hat{y}^T \hat{y}}{y^T y}} = R_o$$

Q (12) 何时 $\bar{y}, \bar{\hat{y}}$ 相等

$$\Leftrightarrow \frac{1}{n} \cdot 1^T Y = \frac{1}{n} 1^T \hat{Y} \Rightarrow 1^T (Y - \hat{Y}) = 0$$

$$\text{又由 } x^T (Y - \hat{Y}) = 0 \Rightarrow \text{当 } X \text{ 包含 } \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ 时, 截距项}$$

Q (13) n_1 个 a 类样本; n_2 个 b 类样本, $n_1 = 2n_2$, 如何应时样本不平衡

$$MSE = \frac{n_1}{2} (y_i - x_i^T \beta_1)^2 + \frac{n_2}{2} (y_i - x_i^T \beta_2)^2$$

① Weighted Least Square: e.g. Let $w_a = \frac{1}{n_1}$ $w_b = \frac{1}{n_2}$
使两 Loss 权重相同

② Undersampling / Oversampling (Copy or SMOTE)

Q114)

$$y \sim (1, x_1), R_1^2 = 0.3$$

$$y \sim (1, x_2), R_2^2 = 0.4$$

$$y \sim (1, x_1, x_2), R_3^2$$

* $R^2 = r_{y,x}^T r_{xx}^{-1} r_{y,x}$ 其中 $r_{y,x} = \begin{pmatrix} r_{y,x_1} \\ r_{y,x_2} \end{pmatrix}$ $r_{xx} = \begin{pmatrix} 1 & r_{x_1x_2} \\ r_{x_1x_2} & 1 \end{pmatrix}$

$$R^2 = \begin{pmatrix} r_{y,x_1} \\ r_{y,x_2} \end{pmatrix}^T \begin{pmatrix} 1 & r_{x_1x_2} \\ r_{x_1x_2} & 1 \end{pmatrix}^{-1} \begin{pmatrix} r_{y,x_1} \\ r_{y,x_2} \end{pmatrix} = \frac{1}{1-r_{x_1x_2}^2} \begin{pmatrix} r_{y,x_1} \\ r_{y,x_2} \end{pmatrix}^T \begin{pmatrix} 1 & -r_{x_1x_2} \\ -r_{x_1x_2} & 1 \end{pmatrix} \begin{pmatrix} r_{y,x_1} \\ r_{y,x_2} \end{pmatrix}$$

$$= \frac{1}{1-r_{x_1x_2}^2} \left(r_{y,x_1}^2 + r_{y,x_2}^2 - 2r_{x_1x_2} r_{y,x_1} r_{y,x_2} \right)$$

本题中: $R^2 = \frac{0.7 - 2\sqrt{0.12} D \cdot r_{x_1x_2}}{1 - r_{x_1x_2}^2}$ 其中, $D = \text{sgn}(r_{y,x_1}) \cdot \text{sgn}(r_{y,x_2}) \in \{+1, -1\}$

$$\Rightarrow R_3^2 > \max(R_1^2, R_2^2)$$

$$R_3^2 \text{ 也 可能 } > 0.7 \text{ "enhancement"}$$

$$\Delta R_3^2 - R_1^2 = \frac{(r_{y,x_2} - \rho r_{y,x_1})^2}{1 - \rho^2} > 0 \quad R_3^2 - R_2^2 > 0 \Rightarrow R_3^2 \geq \max\{R_1^2, R_2^2\}$$

$$\Delta \text{ 当 } \rho = 0: R_3^2 = R_1^2 + R_2^2$$

$$\Delta \text{ 当 } \rho = \pm 1, \text{ 不满足, } R_3^2 = \max\{R_1^2, R_2^2\}$$

Q115) OLS中: X 重尾的问题? Heavy-tail

一般用 kurt 检验 ($k > 3$: 尖峰厚尾).

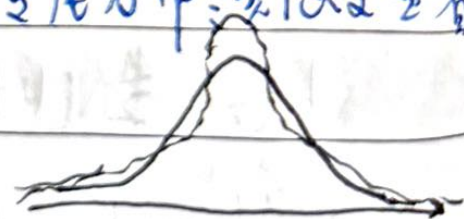
Normal Q-Q plot. (检验 Norm)

$E[X^k]$ 矩阶重尾

$$= \int_{-\infty}^{\infty} x^k p(x) \cdot dx$$

$x = x'$ 时: X^k 为矩阶/损失

对于重尾分布, 刻画更显著



变量 Collinearity 有 (附):

Lasso & Ridge.



当 X 中太多有用特征, 不想完全舍弃 $\rightarrow$ Ridge L2.

* Sum: 当 X 中少量有用特征 + 可解释性 $\rightarrow$ Lasso L1

当 X 有高度共线性 $\rightarrow$ Ridge (Lasso 对多重共线性变量是随机 select)

• elastic net:

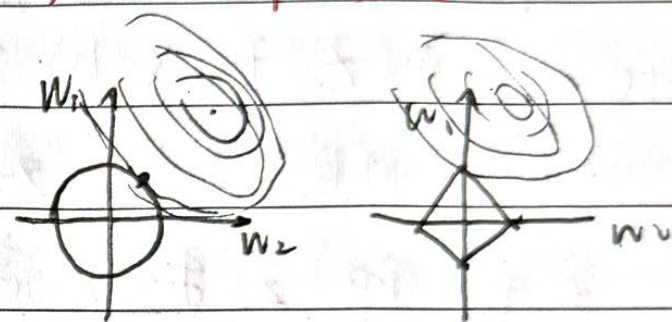
$\sim$ Lasso 有 Sparse solution

设计: 1. 当 $p > n$, Lasso 只能有选出 n 个变量
e-n 设了 λ 限制

2. Lasso 无 grouping effect, 只能任意选

* Ridge L2: $\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2$

Lasso L1: $\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$



Elastic Net: $\min_{\beta} \|y - X\beta\|_2^2 + \lambda (\alpha \|\beta\|_1 + (1-\alpha) \|\beta\|_2^2)$

* 实际应用: Lasso 中通过 soft-thresholding, 把系数很小的设为 0

* 调参: $\lambda \uparrow$, 正则化 penalty 越大.

* 添加新的 x , R^2 怎么变?



$\Rightarrow R^2$ 一定不会下降.

* 原矩阵 $X \in \mathbb{R}^{n \times k}$, 加新变量 $z \in \mathbb{R}^n$.

子空间: $S = \text{span}(X)$ $T = \text{span}(Xz)$

$\Rightarrow$ 有 $S \subseteq T$

记 S 和 T 对应的正交投影为 P_S, P_T

$$e_{old} = y - P_S y \quad e_{new} = y - P_T y$$

因为 $S \subseteq T$ 且正交投影使欧氏距离最小

有 $\|e_{new}\| \leq \|e_{old}\| \Rightarrow SSE_{new} \leq SSE_{old}$

Note: z 和 X 的相关性会影响

但更多是 z 和 y 是否相关

e.g. 即 z 和 X 高度相关:

1. $VIF \uparrow$

2. 如果 y 和 z 相关, R^2 会 $\uparrow$

e.g. 当 z 和 X 完全相关:

$$\text{span}(X) = \text{span}(X, z), R^2 \text{ 不变}$$

* 内生性: Endogeneity: $\text{Cov}(x, u)$ 和

内生性会直接影响 β 估计 + 标准检验不可用

e.g. Omitted Variable Bias:

遗漏变量, 且与已有 X 相关

Note: 遗漏变量和已有变量 X 独立时, 不影响 α 的估计.

Proof: $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$

$$y = \alpha + \beta_1 x_1$$

$$\hat{\beta}_1 = \frac{\text{Cov}(x_1, y)}{\text{Var}(x_1)}$$

$$\text{其中, } \text{Cov}(x_1, y) = \text{Cov}(x_1, \alpha + \beta_1 x_1 + \beta_2 x_2 + \varepsilon) \\ = \beta_1 \text{Var}(x_1) + \beta_2 \text{Cov}(x_1, x_2)$$

$$\text{当 } \text{Cov} = 0 \Rightarrow \text{Cov}(x_1, y) = \beta_1$$

eg. 割的可能原因:

因果关系双向, x, y , mutually causal ~~(eg. Demand)~~



Measurement Error: x & y 测量值有 error

方法: 工具变量, DID, 重新建模.

* Ridge, Lasso, 的 bayes prior?

OLS, L1, L2 在给定 Likelihood Func T , 不同 prior T 的 MAP.

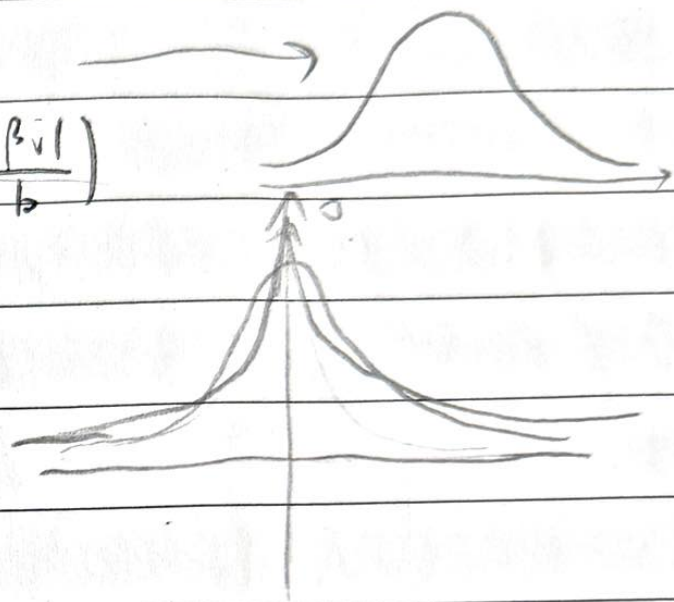
① OLS: β flat prior, uniform distribution

② L2: $\beta \sim N(0, \sigma^2)$ Gaussian prior

③ L1: $\beta \sim \text{Laplace 分布}$ $p(\beta_j) = \frac{1}{2b} \exp(-\frac{|\beta_j|}{b})$

$$f(\beta) = \frac{1}{2a} e^{-\frac{|\beta - \mu|}{a}}$$

Sharpe peak & heavy tails



* VIF: Inflation Factor:

$$VIF_j = \frac{1}{R_j^2}, R_j \text{ from } X_j \sim (1, x_1, \dots, x_{j-1}, x_{j+1}, \dots)$$



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$\frac{S_{xy}}{S_{xx} S_{yy}}$

* $y \sim (1, x)$ - 元 OLS 中: $R^2 = r^2$, $r^2 = \frac{\text{cov}(x, y)^2}{\text{Var}(x) \text{Var}(y)} = \frac{[\sum (x_i - \bar{x})(y_i - \bar{y})]^2}{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}$

Proof: $R^2 = \frac{SSR}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$, 有 $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \Rightarrow \hat{y}_i - \bar{y} = \hat{\beta}_1 (x_i - \bar{x})$

$\Rightarrow R^2 = \hat{\beta}_1^2 \frac{\sum (x_i - \bar{x})^2}{\sum (y_i - \bar{y})^2} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2$

* Let r 是 ρ 的估计: 即使 x, y 独立 $\Leftrightarrow \rho = 0$, 但 r 在样本上依旧有分布.

* 有 $\rho = 0$ 时: $r^2 \sim \text{Beta}(\frac{1}{2}, \frac{n-2}{2})$

* 引理: $\rho = 0$ 时: $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$ 证明 part 1

Δ : $t = \frac{\hat{\beta}_1 - 0}{\text{s.e.}(\hat{\beta}_1)} \sim t_{n-2}$

时: $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = r \cdot \sqrt{\frac{S_{yy}}{S_{xx}}}$

时: $\text{s.e.}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}} = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$

$\Rightarrow t = r \cdot \sqrt{\frac{S_{yy}}{S_{xx}}} \cdot \frac{\sqrt{S_{xx}}}{\hat{\sigma}} = \frac{r \cdot \sqrt{S_{yy}}}{\hat{\sigma}}$

引理: $\hat{\sigma}^2 = \frac{\hat{\epsilon}^T \hat{\epsilon}}{n-k}$, 在 $y \sim (1, x)$ 中: $\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{\sum (\hat{y}_i - y_i)^2}{n-2}$

引理: $SST = SSE + SSR$

时: $t = r \cdot \sqrt{S_{yy}} \cdot \sqrt{\frac{n-2}{SST - SSR}} = r \sqrt{n-2} \cdot \sqrt{\frac{\sum (y_i - \bar{y})^2}{\sum (y_i - \bar{y})^2 - \sum (\hat{y}_i - \bar{y})^2}} = \sqrt{\frac{1}{1-R^2}} = \frac{1}{\sqrt{1-r^2}}$

#

$S_{yy} = SST$

$\sqrt{\frac{SST}{SST - SSR}} = \sqrt{\frac{1}{1 - \frac{SSR}{SST}}} = \sqrt{\frac{1}{1 - R^2}} = \frac{1}{\sqrt{1-r^2}}$

* 证明 part 2:

有 $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$

即: $t^2 = \frac{r^2(n-2)}{(1-r^2)} \sim F_{1, n-2}$

Proof: $t_{n-2} = \frac{z}{\sqrt{\chi^2_{n-2}}}$

$t^2 = \frac{z^2}{(\chi^2_{n-2})} \sim F_{1, n-2}$

* 证明 part 3: r 的分布

$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \Rightarrow r^2 = \frac{t^2}{t^2 + (n-2)}$

由: $t^2 \sim F_{1, n-2}$:

Beta 分布 B 定义为: $B = \frac{d_1 F}{d_1 F + d_2} \sim \text{Beta}(\frac{d_1}{2}, \frac{d_2}{2})$.

$\Rightarrow r^2 \sim \text{Beta}(\frac{1}{2}, \frac{n-2}{2})$.

$\Rightarrow E[r^2] = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{n-2}{2}} = \frac{1}{n-1}$

题中: $n=100, E[r^2] = \frac{1}{99}$

自相关:

eg. 1: 时序数据 2: 截面 spatial autocorr
spatial autocorr



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* 检验:

1. residual 的 ACF, PACF, $ACF = \text{corr}(X_t, X_{t+k})$ $PACF = \text{corr}(X_t, X_{t+k} | X_{t+1}, \dots, X_{t+k-1})$
2. BG 检验, 1) 假设 ε 有 p 阶自相关, 然后 $\{e_t\}$ 与辅助回归
3. Q 检验.

* GLS:

$\Rightarrow$ PW (Prais - Winsen): 假设 $AR(1): \varepsilon_t = \rho \varepsilon_{t-1} + u_t$

有 $\Omega = \begin{pmatrix} 1 & \rho & \dots & \rho^{n-1} \\ \rho & 1 & \dots & \rho^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \dots & 1 \end{pmatrix}$, 则 $\Omega^{-1} = P^T \cdot P$, $P = \begin{pmatrix} \frac{1}{\sqrt{1-\rho^2}} & 0 & \dots & 0 \\ -\rho & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & -\rho & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{1-\rho^2}}$

$$\hat{\beta}_{PW} = (X^T \Omega^{-1} X)^{-1} X^T \Omega^{-1} y, \quad PY = PX\beta + \varepsilon.$$

实际中: ρ 不知道, 先 $e_t \sim (1, e_{t-1})$ 与辅助回归得 $\hat{\rho}$

$\Rightarrow$ 另一种角度: $\varepsilon_t = \rho \varepsilon_{t-1} + u_t$ 已知, 则 λ

$$\text{令 } y_t - \rho y_{t-1} = \beta_0(1-\rho) + \beta_1(x_t - \rho x_{t-1}) + \varepsilon_t.$$

* HAC: heteroskedasticity and Autocorrelation Consistent, stand error / Newy - West.

$$HAC = (X^T X)^{-1} X^T \hat{\Omega} X (X^T X)^{-1}$$

异方差: $\text{Var}(\varepsilon_i | x_i) \neq \sigma^2$
 $= \sigma_i^2$



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White: 修正 OLS
 WLS: 加权 OLS

* White's: 不改变 OLS 的 $\hat{\beta}$, 只修改标准误差计算方式
 Standard Errors

- 不 model the heteroskedasticity explicitly
- observed residuals $\rightarrow$ estimate the variance of $\hat{\beta}$

$$\# \text{Var}(\hat{\beta}) = (X^T X)^{-1} X^T \text{Var}(\varepsilon | X) X (X^T X)^{-1}$$

$$= (X^T X)^{-1} X^T \Omega X (X^T X)^{-1}$$

White $\Rightarrow$ 用 residual 代替 ε_i

$$\hat{\text{Var}}(\hat{\beta}) = (X^T X)^{-1} X^T \begin{pmatrix} \hat{\varepsilon}_1^2 & & \\ & \ddots & \\ & & \hat{\varepsilon}_n^2 \end{pmatrix} X (X^T X)^{-1}$$

使得: White's stand errors 是 consistent 在异方差 (即使 unknown form)

从而: Valid inferences: t, confidence intervals, ...

局限:

1. Small samples, biased downwards. 容易低估 se.

只有当: 大样本故估计: asymptotically consistent estimator

2. 和 WLS 比, 当异方差已知形式, less efficient

* Weighted Least Square:

想法: 对每个样本 weight, (大小和 variance 反比)

$$\# y = X\beta + \varepsilon, \text{Var}(\varepsilon) = \begin{pmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{pmatrix} = \Omega$$

$$\hat{\beta}_{wls} = (X^T W X)^{-1} X^T W y, \quad W = \Omega^{-1} = \begin{pmatrix} \frac{1}{\sigma_1^2} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n^2} \end{pmatrix}$$

$\Rightarrow \hat{\beta}_{wls}$ is BLUE 当 W 正确确定.

$$\text{Var}(\hat{\beta}_{wls}) = (X^T W X)^{-1}$$

$\Rightarrow$ 难点: 很难知道 $\sigma_1^2, \dots, \sigma_n^2$ 的正确估计

想法: $P = \begin{pmatrix} \frac{1}{\sigma_1} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n} \end{pmatrix}$ Transform Matrix $\Rightarrow P^T P = W = \begin{pmatrix} \frac{1}{\sigma_1^2} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n^2} \end{pmatrix}$

$$y = X\beta + \varepsilon \xrightarrow{P} (Py) = (PX)\beta + (P\varepsilon)$$

$$A: \text{Var}(P\varepsilon) = P \text{Var}(\varepsilon) P^T = P \begin{pmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{pmatrix} P^T = I_n.$$

$$\Rightarrow \hat{\beta}_{wls} = (X^T P^T P X)^{-1} X^T P^T P y = (X^T W X)^{-1} X^T W y$$

如何确定 weight?

①: 已知残差的形式 model: e.g. $\text{Var}(u_i | x_i) = \sigma^2 h(x_i) \Rightarrow \text{weight} = \frac{1}{\sigma^2 h(x_i)}$.

②: 不知道 model 形式 ($h(x_i)$),

用 OLS 给 residual 建模. $\Rightarrow$ "Two-step"

$$\Rightarrow \ln(\hat{u}_i^2) = \alpha_0 + \alpha_1 \ln(x_i) + v_i$$

$$\Rightarrow \text{OLS 且得到: } \ln(\hat{\sigma}_i^2) \Rightarrow \hat{\sigma}_i^2 = e^{\hat{\ln(\sigma_i^2)}}, \quad W_i = \frac{1}{\hat{\sigma}_i^2}$$

$\Rightarrow$ Step 2: WLS.

③ 不知道 σ 的 model, 也不想建模:

$\Rightarrow$ 修正 s.e. (Whitís).